DS4420: Machine Learning and Data Mining 2
Dr. Gerber
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Northeastern University

New MovMie: Leveraging Collaborative Filtering and a Multilayer Perceptron Network for Movie Recommendations

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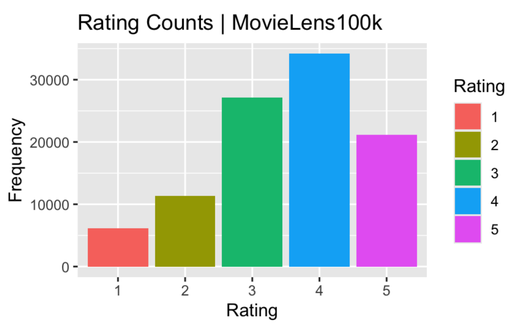
Purpose

- To evaluate collaborative filtering (CF) and multilayer perceptron (MLP) applications to movie recommendations.
- To combine CF and MLP to identify a movie recommendation for the user that they would not try on their own but would likely enjoy.

Background

- Addictive algorithms on video streaming platforms and the like cause echo chambers of information, doom-scrolling or watching, and impact mental health and work efficiency.
- Previous work explores CF and MLP for movie recommendation systems, but this combination is unprecedented in this manner.



Data Overview

- MovieLens 100k dataset
- Rating distribution:
  - {1: 6110, 2:11370, 3:27145, 4:34174, 5:21201}
- Released 4/1998
- CF Data:
  - 1000 users, 1700 movies
- MLP Data:
  - Rating, timestamp, genre, title, age, gender, occupation, zipcode, bias

Methods

User-User Collaborative Filtering (UUCF)
UUCF makes recommendations based on similar users under the assumption that users who liked similar movies in the past will continue to do so in the future.

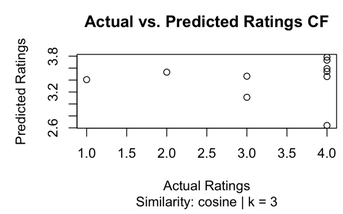
Multilayer Perceptron (MLP)
MLP takes input data and passes it through layers, which learn patterns in data by adjusting weights through backpropagation.

UUCF + MLP Hybrid Model
- Generated datasets of [user | movie | rating] from collaborative filtering and MLP
- Extracted movies the user has not rated, the MLP scores, and the CF scores
- Evaluated scores, returning movie with the highest CF rating

Results

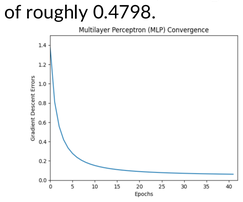
User-User Collaborative Filtering (UUCF)

The best performing hyperparameters were *cosine* for the similarity metric and *3* for the k-value determined through RMSE (1.05), MAE (0.77), and rounded accuracy (0.6) metrics on a test set of 9 masked users.



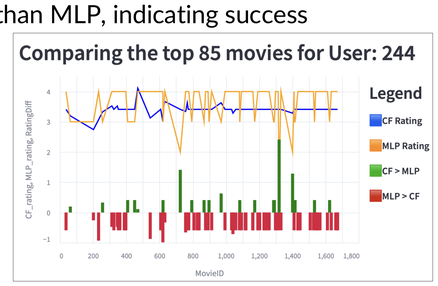
Multilayer Perceptron (MLP)

Utilizing a step size of 0.1, an input layer with 2500 nodes, a single hidden layer with 10 nodes, log loss objective function, and the ReLU activation function, and an output with a single node (movie rating) and the sigmoid activation function, the algorithm produced an accuracy of roughly 0.4798.



UUCF + MLP Hybrid Model

- Preferred and recommended genres differed
- CF score consistently higher than MLP score, with MLP scores low
- For users with preferred and recommended genres with overlap, CF score was still higher than MLP, indicating success



User	Recommended Movie	CF Score	MLP Score	Preferred Genre(s)	Rec. Movie Genre
196	ID: 357 Title: One Flew Over the Cuckoo's Nest (1975)	4.246564	4.0	Comedy	Drama
186	ID: 286 Title: The English Patient (1996)	4.609684	4.0	Thriller	Drama, Romance, War
22	ID: 183 Title: Alien (1979)	4.334328	4.0	Action, Drama	Action, Horror, Sci-Fi, Thriller
244	ID: 483 Title: Casablanca (1942)	4.67168	4.0	Comedy	Drama, Romance, War
166	ID: 228 Title: Star Trek: The Wrath of Khan (1982)	3.899833	3.0	Thriller, Action, Mystery	Action, Adventure, Sci-Fi
298	ID: 192 Title: Raging Bull (1980)	4.451612	4.0	Feel-good movies (i.e. Children's, Comedy, Romance)	Drama
115	ID: 169 Title: The Wrong Trousers (1993)	5.0	4.0	Drama, Action	Animation, Comedy
253	ID: 195 Title: The Terminator (1984)	4.127183	4.0	Attention-grabbing movies (i.e. Action, Adventure)	Action, Sci-Fi, Thriller
305	ID: 23 Title: Taxi Driver (1976)	4.355715	4.0	Drama	Drama, Thriller
6	ID: 654 Title: Chinatown (1974)	4.671806	4.0	Drama	Film-Noir, Mystery, Thriller

Conclusions

- There are trends in the hybrid model's outputs (comedy/feel-good → drama and vice versa).
- Hybrid model CF + MLP offers promising results for suggesting a movie the user would not try on their own assuming they think similarly to MLP
- Hybrid model also indicates potential for the movie to be highly rated due to the recommendation being made on account of similar users' enjoyment

Next Steps

- Test hybrid model on additional data, subsets of this data
- Test hybrid model on newer data with additional characteristics
- Adjust algorithm to apply to short-form videos (such as TikTok, Instagram Reels, etc.)

References

Hou, Y., Xiong, D., Jiang, T., Song, L., & Wang, Q. (2019). Social Media Addiction: Its impact, mediation, and intervention. *Cyberpsychology: Journal of Psychosocial Research on Cyberspace*, *13*(1). https://doi.org/10.5817/cp2019-1-4

Mohamad, A. Y., Harun, S. R., Shahidan, N. A., Nanthaamornphong, A., Mustapha, A., & Wahab, M. H. (2022). Collaborative filtering approach for movie recommendations. *2022 19th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)*, 1–6. https://doi.org/10.1109/ecti-con54298.2022.9795373

Yi, Lei, Shujuan Ji, Lingmei Ren, Rui Su, and Yongquan Liang. "A Nonlinear Feature Fusion-Based Rating Prediction Algorithm in Heterogeneous Network." *IEEE Transactions on Computational Social Systems* 8, no. 3 (January 13, 2021): 728–36. https://doi.org/10.1109/tcss.2020.3046772.

Zhao, Z., Hong, L., Wei, L., Chen, J., Nath, A., Andrews, S., Kumthekar, A., Sathiamoorthy, M., Yi, X., & Chi, E. (2019). Recommending what video to watch next. *Proceedings of the 13th ACM Conference on Recommender Systems*. https://doi.org/10.1145/3298689.3346997